FPGA Implementation of Hardware Accelerator for Real-time Video Image Edge Detection

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***Abstract*—Image edge is considered to be the most important attribute to provide valuable image perception information. At present, video image data is developing towards high resolution and high frame number. The image data processing capacity is huge, so the processing speed is very strict to meet the real-time performance of image data transmission. In this context, we present a method to accelerate the real-time video image edge detection. FPGA is used as the development platform. The real-time edge detection algorithm of image data with 1280x720 resolution and 30 frame/s, combined with median filter, Sobel edge detection algorithm and corrosion expansion algorithm, makes the running time of image processing module shorter. The color image of the video image collected by camera is processed. The HDMI interface shows that the scheme has achieved ideal results in the FPGA hardware platform simulation model, greatly improves the efficiency of the algorithm, and provides a guarantee for the speed and stability of the real-time image processing system.**

Keywords—FPGA; Sobel; Median-filter; Edge Detection

I. INTRODUCTION

The purpose of edge detection is to detect the position of image feature changes to reflect the important changes of the image [1]. Commonly used edge detection algorithms are usually divided into two categories: first derivative edge detection algorithm and second derivative edge detection algorithm. The first derivative: Roberts crossover operator, Sobel operator and kirsch operator respectively. Second derivative: Mar Hildreth operator, LOG operator, Canny operator and Laplacian operator. Sobel operator is one of the most important operators in image edge detection. The principle is relatively simple and the detection effect is good. We can easily implement the algorithm with field programmable gate array (FPGA) [2]. Sobel operator is a first-order difference operator, which judges the edge position of the image by approximating the change of gray value. Through smoothing noise processing, relatively accurate image edge information can be obtained. Although Sobel operator is relatively simple, its efficiency in practical application is also higher than Canny operator. For applications that require high efficiency but do not care about fine texture, Sobel operator is usually the first choice. Due to the video image taken by the camera, there will be a lot of noise in the middle process. However, the traditional Sobel can only give the results of vertical and horizontal edges, and can not detect other gradient directions [3]. Therefore, a large amount of edge information will be lost, which is not conducive to image edge detection [3] [4]. In order to solve this problem, this paper uses 8-direction Sobel to improve the edge strength and meet the requirements of complex edges and high-resolution accuracy.

Edge is an important part of digital image processing. With the development of edge detection technology, the accuracy of edge detection is continuously improved, and the complexity of the program is higher and higher. Many edge detection applications require high real-time performance and low power consumption [5]. Because the operation efficiency of serial execution is too low, the phenomenon of frame loss is common, and takes up more processing time, it is difficult to meet the requirements of speed and accuracy. At present, the field of image processing has entered the white hot stage, which requires the system to have the characteristics of large data processing capacity, high efficiency, high frame rate, good stability and good real-time performance. At present, video image data is developing towards high resolution and high frame number [6]. The amount of image data processing is huge, so the processing speed, image edge and noise are very strict.

Therefore, our contributions are as follows:

1). We use FPGA as the hardware platform of real-time video image edge detection method for hardware accelerator. The data is collected through the camera, combined with median filter, eight-direction Sobel algorithm and corrosion expansion operation method.

2). This design reduces the overall running time of the image processing module. Due to the relatively simple circuit structure of the image processing module in median filtering and Sobel algorithm, corrosion and expansion, the running time is small, so that the camera can realize 1280x720p images, the frame rate can reach 30 frames per second, and display the edge images of high-definition multi frame images through HDMI.

II. FPGA IMPLEMENTATION OF EDGE DETECTION ALGORITHM

A. Hardware Accelerator Structure

Ov5640 image acquisition is our custom IP core, which is responsible for converting Ov5640 data into video stream data. After the image data in YUV format is collected by the Ov5640 camera acquisition module, the data is subjected to Sobel edge detection processing, and then sent to the video in to AXI4 stream IP core. The video stream data is converted into AXI4 stream IP format data stream through the Video in to Axi4 stream IP core, and then converted into AXI4 memory

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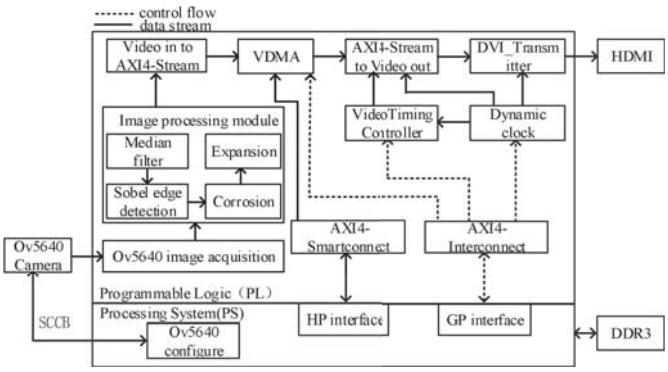

Figure 1. Block diagram of the hardware accelerator structure

map format through the write channel of Video Destination Multiple Access(VDMA), And finally written to DDR memory. VDMA communicates with AXI4 through the AXI4-smartconnect IP core. The HP port is connected to efficiently access DDR3. The video or image data read by VDMA from DDR3 is transmitted to the AXI4 stream to video out IP core [7]. AXI4 stream to video out IP core converts the data in axi4 stream format into the data format of video output under the control of VTC IP core, and finally outputs it to HDMI. The overall architecture of the system is shown in Fig. 1.

B. The Flow Chart of Image Processing

The main design of this design lies in the image processing module, which is mainly composed of median filter, Sobel and corrosion expansion. Because the design adopts Xilinx's zynq platform, according to the design requirements, the algorithm module used in this design needs to be encapsulated into an IP, called image processing module. Its flow chart is shown in the Fig. 2.

C. Median Filter

1)*3x3 Matrix*

There are three ways to generate 3x3 matrix: Ram storage, Fifo storage and shift register storage. This paper uses three shift registers to realize the 3x3 matrix. The schematic diagram is shown in Fig. 3 (b) .

When the system receives the real-time video camera image obtained by continuous data streams, you first need to have storage space to cache the acquisition of data, and data in an image and a serial output into the operation of parallel output[4],In this paper, the bit width of data is defined as 8bit, each line has 1024 data, and two lines of data are registered at the same time. Firstly, the input signal is delayed by one cycle under the action of the enable clock, which can ensure the data synchronization. Secondly, input and instantiate the required three lines of data, through shift storage, you can get the desired 3x3 matrix.

In the shift memory module, two Ram are used to store the data of the first two lines of the image. Ram1 stores the data of the first line (the previous line), Ram0 stores the data of the second line (the previous line), and the input image data is the third line. As shown in Fig. 3(a) .

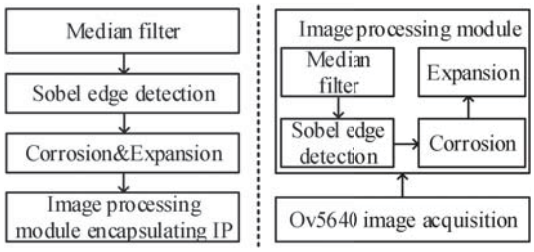

Figure 2. Block diagram of image processing flow chart

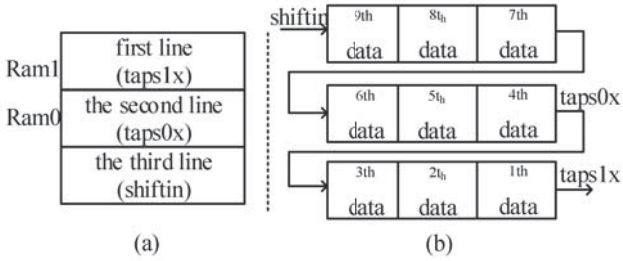

Figure 3. Data flow of shift memory

After reading out the first two lines of image data in the two ram, we need to write the data in Ram0 to Ram1, and then write the new line of image data to Ram0, so as to constantly update the image data in the two ram.

2)*Principle of Median Filter*

In digital image processing, whether the gray image obtained directly or the gray image converted from color image, there is noise, which has a great impact on the image quality. Median filtering is used to remove noise and improve post-processing results. It is widely used in digital image processing because it retains the edge while removing noise [8]. Select the 3x3 matrix here. The template consists of several adjacent pixels. Sort the pixels of the template from small to large, and then replace the original pixel value with the median of the template. The template [10] is shown in Fig. 4.

Firstly, a 3x3 pixel array is generated, and then three pixels in each row are sorted to obtain the maximum, median and minimum values of each row. Then, the sorted matrix is processed, that is, the minimum of three maximum values, the median of three median values and the maximum of three minimum values. Finally, take the median of the three values, and finally get the median of 9 pixels.

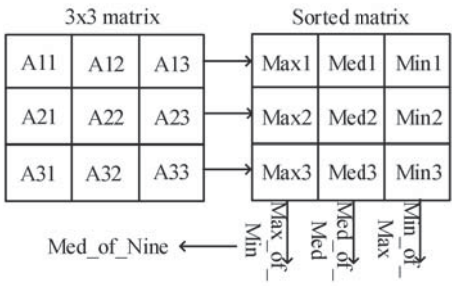

Figure 4. Block diagram of median filtering algorithm

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D. Sobel Algorithm

1) *Traditional Sobel Edge Detection Algorithm*

Sobel operators use vertical and horizontal factors. The principle is to use these two convolution factors to convolute and get two matrices. Then, the square roots of the two corresponding positions of the two matrices are calculated [9]. In addition, the obtained matrix is the gray value, that is, the gradient value of each pixel of the gray image matrix. When it is greater than the input threshold, it is the edge point. The flow chart of Sobel algorithm is shown in the Fig. 5.

In the process of edge detection, the corresponding gray vector or normal vector can be obtained by Sobel operation on any pixel of the image. Sobel is directional and can detect horizontal or vertical edges [9]. Therefore, the operator usually contains two groups of 3x3 matrices in operation [2][9], which are transverse and longitudinal, as shown in Fig. 6, which is Sobel's standard template.

By convoluting the two directional operators with the original image A, the horizontal and vertical differences can be determined[10].

In order to improve the efficiency of image algorithm and realize the advantages of FPGA parallel processing of image data, the implementation process of Sobel algorithm is divided into the following steps.

Step1:Input the original image A;

Step2:Calculate the convolution of the original gray image and Sobel operator in X direction, as shown in formula 1.

$$Gx = A * Sobel_X \tag{1}$$

Step3:Calculate the convolution of the original gray image and Sobel operator in Y direction, as shown in formula 2.

$$Gy = A * Sobel_Y \tag{2}$$

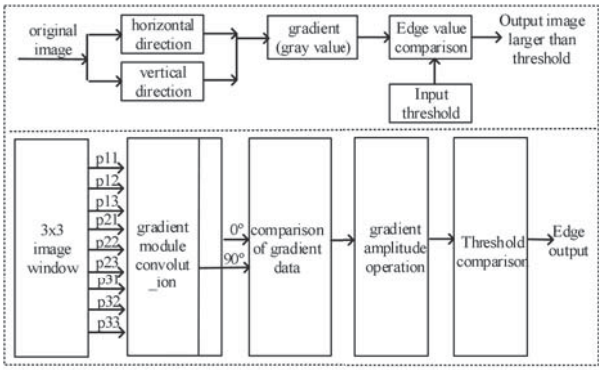

Figure 5. Flow chart of Sobel operator calculation

-1	0	1	1	2	1	p1	p2	p3
-2	0	2	0	0	0	p4	p5	p6
-1	0	1	-1	-2	-1	p7	p8	p9
Gx			Gy			A		

Figure 6. Sobel convolution factor

-1	0	1	0	1	2	1	2	1	2	1	0
-2	0	2	-1	0	1	0	0	0	1	0	-1
-1	0	1	-2	-1	0	-1	-2	-1	0	-1	-2
0º			45º			90º			135º		
1	0	-1	0	-1	-2	-1	-2	-1	-2	-1	0
2	0	-2	1	0	-1	0	0	0	-1	0	1
1	0	-1	2	1	0	1	2	1	0	1	2
180º			225º			270º			315º		

Figure 7. Sobel eight direction operator

Step4:Calculate the size of the gradient, as shown in formula 3.

$$G = \sqrt{Gx^2 + Gy^2} = |Gx| + |Gy| \tag{3}$$

Step5:According to the accuracy requirements, compare with the input threshold to see whether it is greater than the set threshold, and finally determine the edge point.

2) *Eight-Direction Sobel Edge Detection Algorithm*

In this paper, due to the video image taken by the camera, there will be a lot of noise in the middle process. However, the traditional Sobel only gives isotropic results for vertical and horizontal edges, and can not detect other gradient directions. Therefore, a large amount of edge information is lost, which is not conducive to image edge detection[3][4]. In order to improve the edge strength, an improved Sobel template will be used and a diagonal direction detection template will be added[2][4]. It is expanded from the original two directions of 0 ° and 90 ° to eight detection directions[3][4][9], namely 0 °, 45 °, 90 °, 135 °, 180 °, 225 °, 270 ° and 315 °. To improve the effect of edge detection. The detection template in each direction is shown in the Fig. 7.

E. Corrosion Expansion Open Operation

1) *Corrosion*

Corrosion is a process of eliminating boundary points and shrinking the boundary to the inside. It can be used to eliminate small and meaningless objects. Scan every pixel of the image with 3x3 structural elements, and "and" operation with the binary image covered by the structural elements [10]. If both are 1, the pixel in the result image is 1. Otherwise, it is 0. The result will reduce the binary image by one circle.

The implementation of the algorithm can be expressed by the following formula4, that is, the operation of 3x3 pixels.

$$P_{Corrosion} = P11 \& P12 \& P13 \& P21 \& P22 \& P23 \& P31 \& P32 \& P33 \tag{4}$$

In FPGA, in order to convert the speed according to the area, the following formula5 is converted.

$$\begin{aligned} P_{Cor_lin1} &= P11 \& P12 \& P13 \\ P_{Cor_lin2} &= P21 \& P22 \& P23 \\ P_{Cor_lin3} &= P31 \& P32 \& P33 \\ P_{Corrosion} &= P_{Cor_lin3} \& P_{Cor_lin3} \& P_{Cor_lin3} \end{aligned} \tag{5}$$

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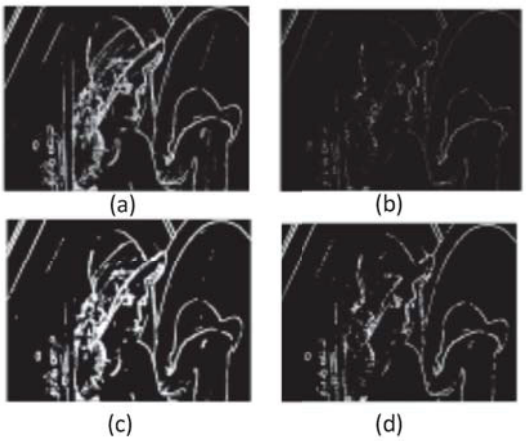

Figure 8. Comparison of the results of corrosion expansion
(a) Sobel (b) Corrosion (c) Expansion

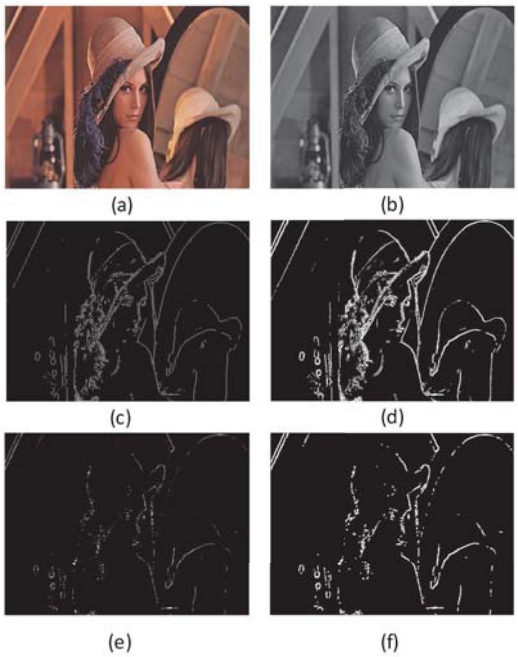

Figure 9. Comparison of algorithms
(a) Original(b)Median filter(c) Traditional sobel (d) 8 Directions sobel
(e) Corrosioned (f) Corrosion before expansion

2) *Expansion*

Expansion is the process of merging all the background points in contact with the object into the object to expand theboundary to the outside. It can be used to fill holes in objects. Each pixel of the image is scanned with 3x3 structural elements, and the binary image covered by the structural elements is used as "or". If both are 0, the pixel in the result image is 0. Otherwise, it is 1[10]. The result will enlarge the binary image by one circle.

The implementation of the algorithm can be expressed by the following formula 6, that is, the operation of 3x3 pixels.

$$P_{Expansion} = P11 | P12 | P13 | P21 | P22 | P23 | P31 | P32 | P33 \qquad (6)$$

In FPGA, in order to convert speed by area, the following formula 7 is converted.

$$\begin{aligned} P_{Exp_lin1} &= P11 | P12 | P13 \\ P_{Exp_lin2} &= P21 | P22 | P23 \\ P_{Exp_lin3} &= P31 | P32 | P33 \\ P_{Expansion} &= P_{Exp_lin1} | P_{Exp_lin2} | P_{Exp_lin3} \end{aligned} \qquad (7)$$

3) *Corrosion Before Expansion*

The process of corrosion before expansion is open operation. In order to eliminate small objects, it is often the first choice to remove the non edge small points around the edge of the image.

As can be seen from the above picture in Fig. 8 (a) is the image after sobel and (b) is the image after corrosion operation. The image is more delicate and suppresses the noise to a great extent(c) Expanded image. It can be seen that the edge of the image becomes thicker and the effect is clearer(d) The image after corrosion and expansion is shown. Compared with the directly obtained Sobel edge, the noise is well suppressed, and the image and edge are clearer.

III. FPGA IMPLEMENTATION

After the simulation, download the generated bit stream to Xilinx development board and call SDK software development tools to better realize the construction and application of face recognition in various fields. After board level testing in MATLAB and development board, the experimental results are shown in Fig. 9.

Through the above simulation results and the images obtained from board level verification, it is confirmed that the FPGA platform can complete the function of the algorithm efficiently and stably.

IV. EXPERIMRNTS ANALYSIS AND RESULTS

The development board used in this design is Xilinx zynq-7020 experimental board. The specific model is xc7z020clg400-2. Ov5640 camera is used to collect external image data, DDR is used to cache the collected data, and HDMI display is used for real-time image display. The purpose of this experiment is to test the running time and frame rate of image processing module, that is, algorithm module. The implementation results are divided into two parts: (1) the realization of resource utilization (2) the comparison of the implementation results of the running time of the image processing module.

A. *Realization of Resource Utilization*

The consumption of resources is often considered in the design. In order to better realize the resource utilization, the utilization of the algorithm module, that is, the utilization of the image processing module and the utilization of the hardware system architecture are well shown in TABLE I. It can be found that in the experimental results of the algorithm process of the image processing module, the resource utilization is 1.84% LUTs, 0.96% FFs 3.64% BRAM and 2.86% IOs. The

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TABLE I. IMPLEMENTATION HARDWARE ARCHITECTURE UTILIZATION

Resource Utilization(%)			
FPGA Component	**Total Available**	**Algorithm Implementation**	**Hardware System Implementation**
LUTs	53200	1.84	14.88
FFs	106400	0.96	11.68
BRAM	140	3.64	7.86
IOs	125	2.86	16.8

resource utilization rate of the hardware system is LUTs with resource utilization rate of 14.88%, FFs with resource utilization rate of 11.68%, BRAM with resource utilization rate of 7.86% and IOs with resource utilization rate of 16.8%.

B. Algorithm Running Time and Frame Rate

In the whole process of image processing algorithm, the image data output by the system is only 0.03s, the frame rate is 30fps and the frequency is 100MHz. For the 1280x720 image data used in the simulation experiment of this design, the median filtering operation is completed in 11.24617ms, the Sobel algorithm module is completed in 11.23929ms, the corrosion is 11.23932ms, the expansion operation is 11.24167ms, and the whole image edge detection and processing module is 44.96645ms. The image data of [3] adopts 128x128 image data. Therefore, the running time of this design is 56.25 times of its running time. Therefore, the running time of each module in this design is generally less than [3]. However, the image data used in [11] is 640x480, that is, the running time of this design is 3 times that of [11]. Therefore, the running time of each module in this design is generally less than [11]. The frequency used in this design is 100MHz, and the frequency used in [3] [11] is 50MHz. After comparison, the experimental results of this design are good and achieve the expected experimental results. The experimental results are shown in TABLE II.

TABLE II. RUNNING TIME OF ALGORITHM MODULE

Time of Algorithm Module			
Algorithm Module	**Proposed**	**Ref.[3]**	**Ref.[11]**
Med(ms)	11.24617	1.2	6.432
Sobel(ms)	11.23929	1.208	6.432
Cor(ms)	11.23932	1.126	-
Exp(ms)	11.24167	1.278	-
Totl-time(ms)	44.96645	4.812	12.864
Time/fra(s)	0.03	0.03	0.03
Frame rate(fps)	30	-	30
Freq (Mhz)	100	50	50
Picture size	1280x720	128x128	640x480

V. CONCLUSION

Edge detection plays an important role in the field of digital image processing and is the basis of many important applications. Taking the hardware acceleration of edge detection algorithm as the background and FPGA as the development platform, this paper processes the real-time edge detection algorithm for the color video image with the resolution of 1280x720 and the number of frames of 30 frames / s collected by ov5640 camera. Combined with median filter Sobel edge detection algorithm and corrosion expansion algorithm, it is displayed through HDMI interface. The experimental results show that the running time on median filter is 11.24617ms, Sobel is 11.23929ms, corrosion module is 11.23932ms, expansion is 11.24167ms, and its resource utilization is 1.84% LUTs, 0.96% FFs, 3.64% Bram and 2.86% IOs. By comparing with similar designs, the running time, speed and resource consumption of image processing module are compared and evaluated. This design can greatly improve the development efficiency.

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